

IoT-Based Smart Drone System for Detecting Crop Insects and Optimizing Pesticide Spraying

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Abstract

Manual crop scouting and blanket pesticide spraying are still the most common practices on many farms, despite their well-known disadvantages, including high labour requirements, excessive chemical use, and health risks for farm workers [1]. To address these challenges, unmanned aerial vehicles (UAVs) equipped with cameras and onboard artificial intelligence have emerged as promising tools for remote sensing and precision agriculture [2], [3].

However, the existing literature remains fragmented. Hardware-focused studies mainly develop spraying mechanisms without incorporating automated disease detection [1]. Deep-learning research achieves high classification accuracy but often overlooks onboard deployment and drone flight constraints [4]. Similarly, flight-planning studies focus on optimizing altitude or energy consumption without integrating pest detection and targeted spraying [5], [6].

This paper proposes an integrated system architecture that combines an altitude-optimized flight and imaging strategy, an onboard or edge-based convolutional neural network (CNN) for real-time crop disease and pest detection, an IoT cloud platform for georeferenced alerts, and a variable-rate spraying system that applies pesticides only to geo-tagged infected areas.

The study reviews and synthesizes twelve peer-reviewed and technical sources to develop a thematic literature map, identify the integration gap that existing studies have not addressed, formulate four measurable research questions, and present a detailed experimental design,

including datasets, deep-learning models, evaluation metrics, and flight trials. The proposed system will be evaluated against conventional manual scouting and blanket pesticide spraying methods.

If successfully validated, the proposed framework is expected to reduce pesticide and water consumption compared with the 500 L/ha manual baseline reported by Ingole et al. [1], shorten the time between disease detection and treatment through onboard inference, and provide a practical, low-cost precision agriculture solution for resource-constrained farming regions.

1. Introduction

Agriculture remains one of the world's most important sectors, providing food, employment, and income for a large portion of the global population. However, crop losses caused by pests and diseases continue to threaten agricultural productivity and food security. It is estimated that global grain production decreases by 10–16% every year because of pests and diseases, while China alone loses approximately 40 million tonnes of grain annually due to these problems [6]. At the same time, traditional crop monitoring is slow and labour-intensive, and blanket pesticide spraying, which is commonly used once an outbreak is suspected, often results in excessive chemical use, environmental pollution, and serious health risks for farm workers, including cancer, hypersensitivity, and asthma [1].

To overcome these challenges, Unmanned Aerial Vehicles (UAVs), commonly known as agricultural drones, have become an effective solution. Compared with satellite-based remote sensing, drones can fly much closer to the crop canopy, are less affected by cloud cover, and can be deployed whenever needed. They can also monitor an area that is 10 to 15 times larger than ground-based scouting within a similar period [2]. Different drone platforms, including multi-rotor, fixed-wing, single-rotor helicopter, and hybrid VTOL systems, offer different levels of endurance, payload capacity, and manoeuvrability, making them suitable for agricultural tasks such as crop monitoring, mapping, and pesticide spraying [2].

Current research has advanced in three main directions. The first focuses on drone hardware and spraying mechanisms, where researchers have developed GPS-guided spraying drones equipped with pump-based nozzle systems and evaluated them against conventional manual spraying

methods [1]. The second focuses on computer vision and deep learning, producing highly accurate crop disease and pest classifiers. For example, a ResNet-50 model achieved 95.2% accuracy on a crop pest and disease dataset collected from Xinjiang farmland, outperforming VGG16, InceptionV3, and MobileNetV2 [4]. However, these models are generally trained and evaluated on high-performance workstation GPUs rather than the lightweight embedded hardware available on drones [3]. The third research direction focuses on IoT and communication systems, where cloud-based architectures connect drones and field sensors to monitoring dashboards and early-warning systems [7], [8], [9]. In parallel, flight-planning research has developed optimization models that improve flight altitude, image resolution, battery efficiency, and wind-aware navigation [5], [6].

Although significant progress has been made in these individual research areas, no reviewed study integrates all four key components—optimized flight planning, onboard edge-based disease detection, IoT-enabled georeferenced alerting, and closed-loop targeted pesticide spraying—into a single evaluated system. This paper addresses that limitation by going beyond a simple literature review. Instead, it proposes a practical and testable system architecture together with a comprehensive experimental design that can evaluate the overall benefits of integrating these technologies compared with the individual approaches reported in previous studies.

1.1 The main objectives of this study are to:

- Synthesize and systematically organize the current state of research on agricultural drones, onboard computer vision, IoT connectivity, and flight-altitude optimization.
- Identify, through the literature synthesis, the key technical gap that prevents existing systems from achieving fully autonomous, low-latency, and targeted crop protection.
- Propose an integrated system architecture along with a corresponding experimental design to evaluate its performance against manual crop scouting and blanket pesticide spraying approaches.
- Quantify the expected trade-offs between detection accuracy and inference latency, flight altitude and coverage, as well as pesticide and water savings versus operational time using evaluation metrics derived from the reviewed literature.

- Position the proposed system within realistic policy, cost, and adoption considerations that are relevant to both smallholder and large-scale farming environments.

2. Literature Review

The twelve sources reviewed in this paper are grouped into four main themes: drone platforms and hardware for agricultural applications, computer vision and deep-learning techniques for crop disease and pest detection, IoT and cloud-based architectures for connectivity and decision support, and flight planning and flight-altitude optimization. Before discussing each theme in detail, **Table 1** provides a summary of the reviewed studies, highlighting their focus, key contributions, and identified limitations.

Table 1: Summary of Reviewed Literature by Focus Area, Contribution, and Gap

Study (Ref.)	Focus / Technology	Key Contribution	Limitation / Gap
[2] Dileep et al., 2020	Drone typology (multirotor, fixed-wing, single-rotor, hybrid VTOL)	Comparative feature/cost/purpose analysis of drone classes for agriculture	Descriptive only; no onboard intelligence or spraying control logic
[5] Abubakar et al., 2023	Flight-altitude optimization model	Closed-form relation between altitude, image resolution (R, P) and battery range (D); optimal altitude ~27.8 m for a 6,000,000 m ² farm	Assumes rectangular, obstacle-free fields; no live disease-detection feedback loop
[3] Basso et al., 2025	Sensors and onboard/edge computer-vision pipeline	Reviews CNN/YOLO/segmentation pipelines (ResNet, MobileNet, U-Net) and	Conceptual review; no field-validated integrated system or

Study (Ref.)	Focus / Technology	Key Contribution	Limitation / Gap
		edge platforms (Jetson, Raspberry Pi)	quantitative spraying savings
[1] Ingole et al., 2024	Hexacopter sprayer with camera-based disease identification	Working prototype; reports 60 L/ha vs. 500 L/ha (manual) water use	Disease identification is manual/visual, not automated; no altitude or IoT optimisation
[12] Viji et al., 2025	ML-based selective spraying via user-uploaded imagery	UI-driven system recommends pesticide type from ML-diagnosed plant condition; targets only affected plants	Diagnosis is image-upload driven, not autonomous in-flight inference; no altitude or IoT layer
[7] Namani & Gonen, 2020	IoT + cloud architecture ("Smart Drone")	Defines sensing–communication–coordination modules and LTE/Wi-Fi cloud pipeline	Architectural proposal; lacks an onboard AI model or empirical accuracy/coverage results
[4] Sun, 2024	ResNet-50 deep-learning pest/disease classifier	95.2% accuracy vs. VGG16/InceptionV3/MobileNetV2; 22 ms inference	Server-class GPU (RTX 3090) training; no discussion of onboard/edge deployment constraints
[10] Ahmed et al., 2018 (IJSDR)	IoT-based automation & dashboard control	Framework for remote greenhouse/field control and sensor data fusion	Pre-2020 architecture; no deep-learning perception layer

Study (Ref.)	Focus / Technology	Key Contribution	Limitation / Gap
[6] Gao et al., 2020	IoT + UAV pest/disease monitoring framework	Solar energy-harvesting IoT nodes + wind-aware UAV flight-path planning; links weather variables to wheat powdery-mildew incidence	Path planning optimises energy, not detection accuracy or spray precision
[11] Ahmed, Marwat, et al., 2024	IoT-based intelligent pest-trap/detection platform	CNN + IoT pipeline for pest counting/identification in the field	Ground-based traps, not aerial; limited spatial coverage of large farms
[8] Dholu & Ghodinde, 2018	Cloud-based IoT for precision agriculture	Sensor-to-cloud architecture (humidity, temperature, soil-moisture nodes)	No UAV component; static ground-sensor network only
[9] Boursianis et al., 2020	IoT + UAV comprehensive survey	Taxonomy of IoT sensor types, protocols, and UAV application scenarios (irrigation, spraying, phenotyping)	Survey-level; does not propose or test an integrated architecture

2.3 Drone Platforms and Spraying Hardware

Dileep et al. [2] provide the most comprehensive classification of agricultural drones among the reviewed studies. They categorize agricultural drones into four main types: multi-rotor, fixed-wing, single-rotor helicopter, and fixed-wing hybrid VTOL. The study also compares each drone type based on its suitability for different agricultural applications, including soil condition assessment, seed planting, pest and disease management, pesticide spraying, crop monitoring,

and livestock surveillance. Among these platforms, multi-rotor drones are identified as the most suitable for spraying and crop surveillance because of their vertical take-off and landing (VTOL) capability and excellent manoeuvrability. Although they have relatively limited endurance and payload capacity, this trade-off is particularly relevant to the proposed system, which requires both precise movement over infected areas and sufficient payload capacity for carrying a pesticide spray tank.

Building on this platform, Ingole et al. [1] developed a hexacopter equipped with a camera gimbal, GPS module, and a piston-driven pump-and-nozzle spraying mechanism. Their study reports that drone-based spraying required only **60 L/ha** of water, compared with **500 L/ha** for the manual application of the same pesticide formulation. This provides the most reliable quantitative baseline among the reviewed studies and serves as the primary comparison point for evaluating the expected resource savings of the proposed system in Section 4. However, disease identification in their system still relies on a human operator who manually reviews the transmitted video. As a result, the overall response time and scalability of the system remain limited.

A related machine-learning-based approach proposed by Viji et al. [12] introduces a selective spraying system with a user interface that analyzes uploaded or captured crop images to identify plant stress or disease and recommend an appropriate pesticide. The system sprays only the affected plants instead of treating the entire field. This approach further supports the idea presented in this paper that onboard disease classification should be directly integrated with variable-rate pesticide spraying, rather than treating disease detection and spraying as two separate processes.

2.4 Computer Vision and Deep Learning for Disease and Pest Recognition

Basso et al. [3] provide a comprehensive review of the sensors and algorithms used for onboard crop monitoring and disease detection. They discuss various imaging technologies, including optical, multispectral, and thermal cameras such as the Parrot Sequoia+ and OAK series, as well as vegetation indices like NDVI and GNDVI. The study also reviews several detection and segmentation techniques, including the Hough Transform for crop-row detection, U-Net and DeepLab for image segmentation, and YOLO, SSD, and Faster R-CNN for object detection.

Importantly, the authors emphasize that real-time, low-latency decision-making in the field depends on onboard processing using lightweight deep-learning models, such as MobileNet and SqueezeNet, deployed on embedded platforms like NVIDIA Jetson or Raspberry Pi, rather than relying on post-flight image analysis.

Sun [4] presents an experimentally validated example of the disease detection component. In the study, a ResNet-50 model was fine-tuned using an augmented dataset of strawberry and apple pest and disease images, achieving 95.2% classification accuracy with an inference time of 22 ms. This performance surpassed VGG16 (91.3%), InceptionV3 (93.7%), and MobileNetV2 (90.5%) on the same dataset. However, the higher accuracy of ResNet-50 comes with a larger model size, containing 25.6 million parameters compared with 3.4 million parameters for MobileNetV2. This highlights the trade-off between detection accuracy and computational efficiency, which is a key design consideration for onboard drone systems, where processing power and battery capacity are significantly more limited than the RTX 3090 workstation used in the study.

Ahmed et al. [11] address the same crop recognition problem from a different perspective by developing a ground-based IoT pest-trap system that combines CNN-based classification with IoT connectivity for real-time pest monitoring and alerts. Although their platform is stationary rather than drone-based, the study demonstrates that GAN-based data augmentation and continual-learning techniques can significantly reduce the amount of labelled training data required. These findings are directly applicable to drone-based disease detection systems, which must accurately recognize different crop types and disease stages without relying on excessively large field-collected datasets.

2.5 IoT and Cloud Architectures for Connectivity and Decision Support

Namani and Gonen [7] propose a three-module "Smart Drone" architecture consisting of sensing, communication, and coordination modules. In their system, a fixed-wing UAV equipped with a 4G/LTE modem transmits georeferenced images to a cloud data center, where the data are used for mission planning, sensor-data mosaicking, and farmer-facing dashboards. Dholu and Ghodinde [8] present a complementary cloud-based IoT architecture that focuses on ground sensors for monitoring humidity, temperature, and soil moisture. Their work highlights the

broader IoT ecosystem within which a drone-based crop detection system should operate, rather than functioning as a standalone solution.

Boursianis et al. [9] provide one of the most comprehensive reviews of IoT and UAV integration in smart agriculture. Their survey covers a wide range of sensor technologies, communication protocols, including Wi-Fi, ZigBee, LoRa, and cellular networks, as well as agricultural applications such as irrigation, fertilization, pesticide spraying, weed management, and field phenotyping. The study confirms that integrating IoT with UAVs has become an established approach in smart farming. However, consistent with the research gap identified in Section 1, the survey does not propose or evaluate a complete closed-loop system that combines disease detection with targeted pesticide application.

Gao et al. [6] contribute two key components that are directly relevant to the proposed methodology. First, they introduce a solar-powered, angle-tracking, energy-harvesting IoT node that extends the operating time of unattended field sensors. Second, they develop a wind-aware UAV flight-path planning model that improves flight endurance by taking advantage of downwind routes. In addition, their empirical analysis of wheat fields in the Yangtze River Zone shows that powdery mildew occurrence is closely associated with temperatures between 14–16°C and relative humidity levels above 65%. These findings can be used to schedule scouting flights during high-risk weather conditions instead of relying on fixed monitoring schedules.

2.6 Flight-Altitude and Path Optimisation

Abubakar et al. [5] present a mathematical model that explains the trade-off between flight altitude, image spatial resolution, and battery-limited flight distance for rectangular agricultural fields. Their model determines three important flight altitudes: a minimum altitude constrained by the drone's battery range, a maximum altitude determined by the minimum acceptable image resolution, and an optimal altitude that minimizes image coverage outside the farm boundary. Using a 4,000 m × 1,500 m farm as an example, together with a 4K camera, an 84° field of view, and a 150,000 m single-charge flight range, the study calculates a minimum altitude of 22.21 m, a maximum altitude of 35.54 m, and an optimal altitude of 27.76 m. The results also show that flying at the maximum altitude can reduce travel distance by up to 33%, while flying at the

minimum altitude can improve image resolution by up to 60% above the minimum required level. Among the reviewed studies, this altitude–resolution–endurance model provides the most practical quantitative framework for planning the imaging phase of the proposed system, as described in Section 3.

Overall, the four themes discussed in the literature demonstrate that significant progress has been made in individual research areas, but these technologies remain architecturally disconnected. Research has produced effective drone platforms and spraying hardware (Section 2.1), highly accurate but computationally demanding deep-learning models (Section 2.2), established IoT and cloud-based connectivity frameworks (Section 2.3), and well-defined flight-altitude optimization models (Section 2.4). However, none of the reviewed studies integrates all four components into a single closed-loop system. This overall synthesis provides the foundation for the integrated system architecture and experimental design proposed in Sections 3 and 4.

2.5 Research Question Development

2.5.1 Research Gap

Although many studies have contributed to smart agriculture, several important limitations still remain.

2.5.1.1 Lack of an integrated smart drone system

Most existing studies focus on only one or two parts of the agricultural process, such as drone spraying, crop disease detection, IoT monitoring, or flight planning. However, very few studies combine all these components into a single closed loop system that can detect crop insects, identify infected locations, communicate through IoT, and perform targeted pesticide spraying automatically.

2.5.1.2 Limited real time onboard decision making

Several studies achieve high accuracy in crop disease or pest detection using deep learning models. However, many of these models are developed and tested on powerful computers instead

of onboard drone devices. As a result, their suitability for real time detection on drones with limited computing resources is still not fully addressed.

2.5.1.3 Weak integration between detection and pesticide spraying

Some drone based systems successfully perform pesticide spraying, while other studies focus on crop disease or pest detection. However, these tasks are often treated as separate processes. Existing work provides limited support for automatically connecting disease detection with selective pesticide spraying based on the detected infected locations.

2.5.1.4 Limited use of IoT for complete field operations

IoT has been widely used for monitoring environmental conditions, collecting sensor data, and supporting cloud based agriculture. However, most studies do not fully integrate IoT communication with drone based detection, geo tagged infection mapping, real time farmer alerts, and post treatment verification in one complete workflow.

2.5.1.5 Flight optimisation is rarely connected with intelligent pest management

Research on drone flight planning mainly focuses on improving image quality, flight coverage, or battery efficiency. However, these optimisation techniques are rarely combined with real time crop insect detection and targeted pesticide spraying to improve both operational efficiency and sustainable pesticide use.

2.5.2 Research Questions (RQs)

The following research questions guide the proposed study. Each question is directly aligned with the project objectives, implemented methodology, and evaluation strategy.

RQ1: How effectively can an IoT based smart drone system integrate real time crop pest detection, georeferenced infection mapping, and targeted pesticide spraying within a single operational workflow?

This research question supports the primary objective of developing an integrated closed loop system. It will be evaluated by implementing the complete system architecture described in Section 3 and verifying the successful interaction among the sensing, edge computing, IoT communication, mapping, and spraying modules.

RQ2: Which deep learning model provides the best balance between detection accuracy and computational efficiency for onboard crop pest and disease recognition on resource constrained drone platforms?

This question supports the objective of selecting a suitable onboard classifier. It will be addressed by experimentally comparing the ResNet 50 model and a lightweight MobileNet based model using classification accuracy, inference time, and computational resource requirements as evaluation metrics.

RQ3: How does altitude optimized flight planning influence image acquisition quality and operational efficiency for agricultural monitoring missions?

This question supports the objective of improving drone mission efficiency. It will be investigated by applying the altitude optimization model proposed by Abubakar et al. and evaluating image coverage, spatial resolution, and estimated flight efficiency under different flight altitudes.

RQ4: To what extent can IoT enabled georeferenced selective pesticide spraying reduce unnecessary pesticide application compared with conventional full field spraying?

This question supports the objective of promoting sustainable pesticide management. It will be evaluated by comparing the proportion of treated areas, estimated pesticide usage, and spraying coverage between selective spraying based on detected infection locations and conventional uniform spraying approaches

3. Methodology

This section specifies the proposed system architecture, its constituent modules, and the design rationale linking each module to the literature reviewed in Section 2.

3.1 System Architecture Overview

The proposed system (Figure 1) is designed as a closed-loop architecture consisting of six interconnected modules that work together, rather than as a simple one-way processing pipeline.

1. **UAV Sensing Layer:** A multirotor UAV equipped with an RGB or multispectral camera, GPS, and a barometer collects crop images and environmental data during flight.
2. **Flight Planning and Altitude Optimization:** Before each mission, an altitude and path-optimization module applies the flight-altitude model proposed by Abubakar et al. [5]. The module determines the optimal flight plan based on the dimensions of the target field and the specifications of the onboard camera.
3. **Onboard Edge Computing:** An embedded computing platform, such as NVIDIA Jetson or Raspberry Pi, following the approach of Basso et al. [3], runs a lightweight CNN model to perform real-time crop disease and pest classification during flight.
4. **Georeferenced Infection Mapping:** Every positive disease or pest detection is automatically timestamped and geo-tagged to create an accurate map of infected locations within the field.
5. **IoT Communication Layer:** An IoT gateway using LoRa or 4G connectivity, based on the architectures proposed by Namani and Gonen [7] and Boursianis et al. [9], transmits alerts to a farmer dashboard while storing flight and detection records in the cloud.
6. **Targeted Spraying Module:** A variable-rate spraying system, based on the pump-and-nozzle mechanism developed by Ingole et al. [1], applies pesticides only to the geo-tagged infected areas, reducing unnecessary chemical usage.

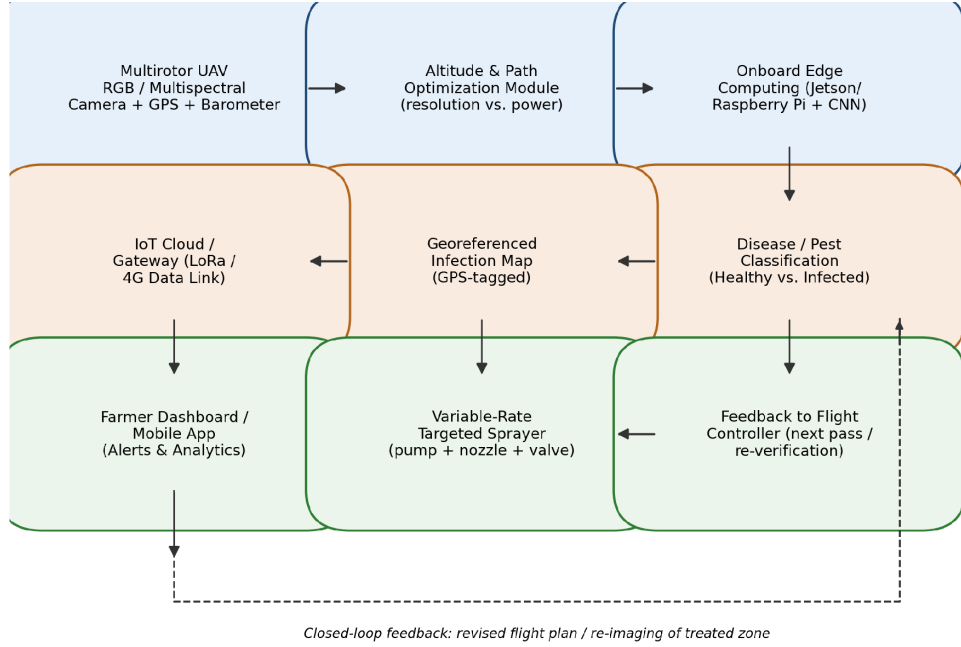


Figure 1: Proposed closed-loop system architecture integrating altitude-optimised flight planning, onboard disease classification, IoT alerting, and targeted spraying.

3.2 Imaging and Flight Planning

Before each mission, the field length (**M**) and width (**N**) are entered along with the onboard camera's resolution (**R**) and field of view (**θ**). Using **Equations (1)–(9)** proposed by Abubakar et al. [5], the system calculates the minimum flight altitude (**h_d**), which is limited by the battery range (**D**), and the maximum flight altitude (**h_p**), which is determined by the minimum acceptable spatial resolution (**P**). Based on these values, the optimal flight altitude (**h_o**) is selected to minimize image over-coverage beyond the boundaries of the target field.

In addition to flight planning, mission scheduling takes into account the temperature and humidity risk conditions identified by Gao et al. [6]. Specifically, temperatures between **14–16°C** and relative humidity levels above **65%**, which are associated with wheat powdery mildew, are used as an example of a disease-risk trigger. As a result, scouting missions can be scheduled more frequently during high-risk weather conditions instead of following a fixed monitoring schedule.

3.3 Onboard Disease and Pest Classification

Two candidate deep-learning models, introduced in Section 2.2, are selected for empirical comparison: ResNet-50, which offers higher classification accuracy but has a larger computational footprint following Sun [4], and a lightweight MobileNet-based model, which provides lower computational requirements at the cost of slightly reduced accuracy, following Basso et al. [3].

Both models are fine-tuned using an augmented image dataset that combines publicly available crop disease datasets with locally collected field images. The data augmentation techniques validated by Sun [4], including rotation, horizontal and vertical flipping, scaling, and random cropping, are applied to increase dataset diversity. These techniques help compensate for the relatively small number of field-collected images while also reducing the class imbalance problem identified for embedded drone deployments [11].

3.4 IoT Connectivity and Targeted Spraying

Each positive detection is automatically timestamped, geo-tagged using the UAV's GPS coordinates, and transmitted through the IoT gateway to a cloud database. This communication follows the two-layer architecture (long-range gateway and local sensor) proposed by Gao et al. [6], together with the cloud dashboard and alerting framework introduced by Namani and Gonen [7].

During the subsequent low-altitude treatment mission, the flight controller revisits only the previously geo-tagged locations. The variable-rate sprayer, based on the pump-and-nozzle mechanism developed by Ingole et al. [1], is activated only within a predefined radius around each tagged point. This targeted spraying approach completes the closed-loop workflow by returning control to the flight-planning module for post-treatment verification, as illustrated in Figure 1.

4. Experimental Design Plan

This section translates the architecture of Section 3 into a testable experimental plan, structured around the four research objectives stated in Section 1.1.

4.1 Dataset Construction and Ground Truth

Field imagery will be collected for at least two economically significant crops, following the multi-institution, multi-season collection protocol used by Sun [4] to ensure coverage of different disease stages, lighting conditions, and viewing angles. Images will be labelled by agricultural extension experts into healthy/infected/pest-damaged categories consistent with the crop-wise disease tables reported by Ingole et al. [1], and split into training, validation, and test sets using the 70%, 15%, and 15% protocol adopted by Sun [4].

4.2 Model Training and Onboard Benchmarking

The ResNet-50 and MobileNet-class classifiers described in Section 3.3 will be trained under identical hyperparameters (Adam optimiser, cross-entropy loss, learning-rate scheduling) following the configuration reported by Sun [4], then benchmarked for accuracy, parameter count, and inference latency both on a workstation GPU and on the target embedded board (Jetson/Raspberry Pi), extending the comparison in Sun [4] which reports only workstation-class results to the resource-constrained setting emphasised by Basso et al. [3].

4.3 Flight-Altitude Validation

For each test field, the altitude model of Abubakar et al. [5] will be used to compute h_d , h_p , and h_o , and three imaging passes will be flown at these altitudes to empirically verify the predicted trade-off between image resolution (input quality for the classifiers of 4.2) and classification accuracy, and between altitude and battery consumption/flight distance.

4.4 Comparative Field Trial

The complete system will be trialled against two baselines drawn directly from the literature: (a) manual scouting with blanket spraying at 500 L/ha [1], and (b) drone-based spraying with human-in-the-loop diagnosis at 60 L/ha [1]. The metrics recorded for all three conditions are: pesticide/water volume per hectare, detection-to-treatment latency, percentage of infected area

correctly treated, percentage of healthy area incorrectly treated (false-positive spraying), total flight/operation time, and classifier accuracy at the altitude selected in 4.3.

4.5 IoT Reliability Testing

The LoRa/4G alerting pipeline will be stress-tested for message-delivery latency and packet loss under the two-layer gateway architecture described by Gao et al. [6], across varying gateway-to-drone distances, to determine whether the communication layer can support the near-real-time alerting required by the closed loop in Figure 1.

5. Discussion

The experimental design in Section 4 is expected to show that onboard-deployable lightweight classifiers trade a measurable amount of accuracy for a large reduction in inference latency relative to server-class models, the optimal altitude predicted analytically by Abubakar et al. [5] is also the altitude that empirically maximises classification accuracy per unit of flight time, and closing the detection spraying loop reduces both pesticide volume and treatment latency relative to the human-in-the-loop baseline of Ingole et al. [1].

5.1 Accuracy–Latency Trade-off in Onboard Classification

The gap between Sun's [4] workstation-class 95.2% accuracy and the expected accuracy of an embedded-friendly model is the central engineering trade-off of this proposal. Because MobileNetV2 already trails ResNet-50 by only 4.7 percentage points at roughly one-seventh the parameter count [4], it is plausible that an onboard classifier can retain most of the diagnostic value of the larger model while meeting the real-time constraint identified by Basso et al. [3] as the key bottleneck for field deployment.

5.2 Altitude as a Shared Constraint on Resolution and Energy

Because the same altitude parameter simultaneously governs image resolution (input to the classifier) and flight endurance (input to mission coverage), the altitude-optimisation model of Abubakar et al. [5] cannot be tuned for detection accuracy alone; Section 4.3 is designed specifically to reveal whether the analytically "optimal" altitude h_o also coincides with the empirically best classification altitude, or whether a secondary trade-off must be introduced.

5.3 Expected Reduction in Chemical and Water Use

Using the Ingole et al. [1] baselines (500 L/ha manual vs. 60 L/ha human-diagnosed drone spraying) as reference points, the proposed automated, geo-targeted approach is expected to approach or improve on the 60 L/ha figure by reducing both the diagnosis lag (which currently causes over-spraying of buffer zones around suspected infection) and operator fatigue-related error, consistent with the general finding that IoT/UAV integration enables more efficient resource use and earlier problem detection [9].

5.4 Communication Reliability as a Limiting Factor

The closed loop in Figure 1 is only as fast as its weakest communication link; the Gao et al. [6] finding that LoRa can reach 15 km versus roughly 100 m for ZigBee-class radios suggests that a LoRa-based gateway is preferable for the large, often infrastructure-poor farms this system targets, but the two-layer TVWS/LoRa architecture they propose has not been tested under the higher message frequency a real-time spraying alert system would generate, which the trial in Section 4.5 is designed to quantify.

6. Policy, Technology, and Innovation Pathways

Beyond the technical design, the reviewed literature repeatedly identifies non-technical barriers that determine whether a system such as the one proposed here can actually be adopted at scale.

6.1 Cost and Accessibility for Smallholder Farms

Dileep et al. [2] report that agricultural drone prices range from roughly AUD 5,000–6,000 for multi-rotor platforms to AUD 25,000–300,000 for fixed-wing and single-rotor helicopter systems, placing the higher-end platforms out of reach for smallholder farmers. A realistic adoption pathway therefore favours the multi-rotor configuration used in this proposal, paired with low-cost embedded compute (Raspberry Pi, per Basso et al. [3]) rather than the most accurate but most expensive sensing/compute combination.

6.2 Connectivity Infrastructure Policy

Namani and Gonen [7] and Gao et al. [6] both depend on cellular (4G/LTE) or unlicensed-spectrum (LoRa, TVWS) connectivity that is unevenly available in rural regions;

national spectrum policy that allocates TV white-space or agricultural IoT bands, as already implemented in parts of China [6], is therefore a direct enabler of the IoT layer proposed in Section 3.4.

6.3 Data Governance and Farmer Trust

Because the system stores georeferenced field-health data in a cloud data centre [7], clear data-ownership and privacy policy specifying that farm-level data belongs to the farmer and is not resold or used to disadvantage them commercially is a prerequisite for the trust needed for adoption, a concern echoed in Boursianis et al. [9] regarding IoT data flows in smart farming generally.

6.4 Regulatory Pathways for Autonomous Spraying

Most reviewed systems assume a human operator supervises the drone [1], [7]; a shift toward the fully autonomous, closed-loop spraying proposed in Section 3.4 will require aviation and pesticide-application regulators to define certification pathways for autonomous agricultural UAVs, analogous to the pesticide-formulation-specific application guidance already tabulated by crop and disease in Ingole et al. [1].

6.5 Innovation Pathway: From Detection to Decision Support

The trajectory visible across the reviewed sources from static ground sensors [8] to mobile aerial sensing [2] to onboard AI [3], [4] to closed-loop treatment (this proposal) suggests the next innovation step beyond the present system is predictive rather than reactive: using the weather–disease correlations established by Gao et al. [6] to schedule pre-emptive scouting flights, moving the system from disease detection toward disease forecasting.

7. Conclusion

This paper reviewed twelve studies covering agricultural drone hardware, onboard computer vision, IoT connectivity, and flight-altitude optimization. The review showed that although these four components have each been well developed individually, none of the reviewed studies integrates them into a single, fully evaluated closed-loop system.

Based on this literature synthesis, the paper proposed an integrated system architecture that combines altitude-optimized flight planning, onboard deep-learning-based disease classification, IoT-enabled georeferenced alerting, and targeted variable-rate pesticide spraying. In addition, a detailed experimental design was presented, including dataset construction, onboard model benchmarking, flight-altitude validation, comparative field trials, and IoT reliability testing. This experimental framework is intended to evaluate the proposed system against the manual and semi-automated approaches reported in the existing literature.

Beyond its technical contribution, the paper also considered important practical factors that influence real-world adoption, including cost, connectivity, data governance, and regulatory requirements. These considerations are essential for determining whether integrated precision agriculture systems can move beyond research prototypes and become practical solutions for both smallholder and large-scale farming.

Future research should focus on conducting the experimental trials proposed in Section 4 and reporting their empirical findings. It should also further investigate the trade-offs between detection accuracy and inference latency, as well as between flight altitude and image resolution, using real field data instead of analytical projections alone.

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